

Arthur Cayley

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Contents

607. A Memoir on Prepotentials (continued)

[Continued from earlier pages of the same paper. Articles 96–131.]

[p. 381] 96. The positive root θ of the equation

$$\frac{a^2}{f^2 + \theta} + \frac{c^2}{h^2 + \theta} - 1 + \frac{e^2}{\theta} = 0$$

has certain properties which connect themselves with the function

$$\Theta_0 = \theta^{q-1} \{(\theta + f^2) \dots (\theta + h^2)\}^{-\frac{1}{2}}.$$

We have, the accents denoting differentiations in regard to θ ,

$$J' \frac{d\theta}{da} = \frac{2a}{\theta + f^2}, \quad J' \frac{d\theta}{de} = -\frac{2e}{\theta} J'; \quad \text{and} \quad \frac{d\theta}{da} = \frac{1}{J'} \cdot \frac{2a}{\theta + f^2},$$

where

$$J' = \frac{a^2}{(f^2 + \theta)^2} + \dots + \frac{c^2}{(h^2 + \theta)^2} + \frac{e^2}{\theta^2};$$

and we have the like formulae for $\frac{d\theta}{dc}, \frac{d\theta}{de}$.

We deduce

$$\frac{a}{\theta + f^2} \frac{d\theta}{da} + \dots + \frac{c}{\theta + h^2} \frac{d\theta}{dc} + \frac{e}{\theta} \frac{d\theta}{de} = \frac{2}{J'},$$

and to this we may join, η being arbitrary,

$$\frac{a}{\theta + \eta + f^2} \frac{d\theta}{da} + \dots + \frac{c}{\theta + \eta + h^2} \frac{d\theta}{dc} + \frac{e}{\theta + \eta} \frac{d\theta}{de} = \frac{2}{J'} \left[1 - \left\{ \frac{a^2 \eta}{(\theta + f^2)(\theta + \eta + f^2)} + \dots + \frac{e^2 \eta}{\theta(\theta + \eta)} \right\} \right].$$

Again, defining $\nabla_a \theta$ and $\square \theta$ as immediately appears, we have

$$\nabla_a \theta = \left(\frac{d\theta}{da} \right)^2 + \dots + \left(\frac{d\theta}{dc} \right)^2, \quad = \frac{4}{J'}, \quad \square \theta = \frac{8a^2}{J'(\theta + f^2)^2};$$

and passing to the second differential coefficients, we have

$$\frac{d^2 \theta}{da^2} = \frac{2}{J'(\theta + f^2)} - \frac{8a^2}{J'^2(\theta + f^2)^2} - \frac{4a^2 J''}{J'^3(\theta + f^2)^2},$$

where

$$J'' = -\frac{2a^2}{(f^2 + \theta)^3} - \dots - \frac{2c^2}{(h^2 + \theta)^3} - \frac{2e^2}{\theta^3},$$

and the like formulae for $\frac{d^2 \theta}{dc^2}, \frac{d^2 \theta}{de^2}$. Joining to these $\frac{1}{e} \frac{d\theta}{de} = \frac{2}{J'} \cdot \frac{2q+1}{\theta} \frac{d\theta}{de}$, we obtain

$$\begin{aligned} \square \theta &= \left(\frac{d^2 \theta}{da^2} + \dots + \frac{d^2 \theta}{dc^2} + \frac{d^2 \theta}{de^2} + \frac{2q+1}{e} \frac{d\theta}{de} \right) \\ &= \frac{2}{J'} \left[\frac{1}{\theta + f^2} + \dots + \frac{1 + (2q+1)}{\theta} \right] - \frac{2}{J'^2} \left[\frac{2a^2}{(\theta + f^2)^2} + \dots + \frac{2e^2}{\theta^2} \right] - \frac{J''}{J'^3} (4a^2 \dots). \end{aligned}$$

[p. 382] where the last two terms destroy each other; observing that we have

$$\frac{1}{e^2} \cdot \frac{d}{de} (e^{2q+1}) = \frac{2q+1}{e^2} \cdot e^{2q} \cdot \frac{1}{e} \cdot e = \frac{2q+1}{e^2} \cdot e^{2q-2} \cdot e^2;$$

the result is

$$\square \theta = \frac{2}{J'} \left(\frac{2q+2}{\theta} - \frac{2\theta'}{\theta} \right) + \frac{4\theta'}{\theta J'^2}.$$

97. First example, $x^2 = a^2 + \dots + c^2$, and θ the positive root of $\frac{x^2}{f^2 + \theta} + \frac{e^2}{\theta} = 1$.

V is assumed $= J \int_{\theta}^{\infty} t^{q-1} (t + f^2)^{-\frac{1}{2}s} dt$, where $q + 1$ is positive.

I do not work the example out; it corresponds step by step with, and is hardly more simple than, the next example, which relates to the ellipsoid. The result is

$$\rho = (\sqrt{\pi})^s \Gamma(q + 1)^{-1} \left[\left(1 - \frac{x^2}{f^2} - \frac{e^2}{f^2} \right)^q \right], \quad \text{if } \frac{x^2}{f^2} + \frac{e^2}{f^2} < 1;$$

hence the integral

$$\int \frac{\left\{ 1 - \frac{x^2}{f^2} - \frac{e^2}{f^2} \right\}^q dx \dots dz}{\left\{ (a - x)^2 + \dots + (c - z)^2 + e^2 \right\}^{\frac{1}{2}s + q}},$$

taken over the sphere $a^2 + \dots + z^2 = f^2$,

$$= \frac{(\sqrt{\pi})^s \Gamma(q + 1)}{\Gamma(\frac{1}{2}s + q + 1)} f^s \int_{\theta}^{\infty} t^{q-1} (t + f^2)^{-\frac{1}{2}s} dt.$$

98. Second example, θ the positive root of $\frac{a^2}{f^2 + \theta} + \dots + \frac{c^2}{h^2 + \theta} + \frac{e^2}{\theta} = 1$, $q + 1$ positive.

Consider here the function

$$V = \int_{\theta}^{\infty} t^{q-1} \{ (t + f^2) \dots (t + h^2) \}^{-\frac{1}{2}} dt;$$

this satisfies the prepotential equation. We have in fact

$$\frac{dV}{da} = -\Theta_0 \frac{d\theta}{da}, \quad \frac{d^2V}{da^2} = -\frac{d\Theta_0}{d\theta} \left(\frac{d\theta}{da} \right)^2 - \Theta_0 \frac{d^2\theta}{da^2},$$

with the like expressions for $\frac{d^2V}{dc^2}$, $\frac{d^2V}{de^2}$; also

$$\frac{2q + 1}{e} \frac{dV}{de} = -\Theta_0 \frac{2q + 1}{e} \frac{d\theta}{de}.$$

Hence

$$\square V = -\Theta_0 \square \theta - \theta' \nabla_a \theta,$$

[p. 383] or, substituting for $\square \theta$ and $\nabla_a \theta$ their values, this is

$$= \Theta_0 \left\{ -\frac{2}{J'} \left(\frac{2q + 2}{\theta} - \frac{2\theta'}{\theta} \right) - \frac{4\theta'}{\theta J'^2} \right\} - \frac{4\theta'}{J'} = 0.$$

Moreover V does not become infinite for any values of $(a, \dots, c, e)$, e not $= 0$; and it vanishes for points at ∞ . And not only so, but for indefinitely large values of any of the coordinates $(a, \dots, c, e)$ it reduces itself to a numerical multiple of $(a^2 + \dots + c^2 + e^2)^{-\frac{1}{2}s + q}$; in fact, in this case θ is indefinitely large, $= a^2 + \dots + c^2 + e^2$. Consequently throughout the integral, t is indefinitely large, and we may therefore write

$$V = \int_{\theta}^{\infty} t^{q-1} t^{-\frac{1}{2}s} dt = \frac{-1}{\frac{1}{2}s - q} [t^{-\frac{1}{2}s + q}]_{\theta}^{\infty} = \frac{1}{\frac{1}{2}s - q} \theta^{-\frac{1}{2}s + q};$$

that is,

$$V = \frac{1}{\frac{1}{2}s - q} (a^2 + \dots + c^2 + e^2)^{-\frac{1}{2}s + q}.$$

The conditions of the theorem are thus satisfied, and we have for ρ either of the formulae

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{(\sqrt{\pi})^s \Gamma(q)} e^{2q} W, \quad \rho = \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q + 1)} \frac{1}{e^{2q+1}} \frac{dW}{de};$$

in the former of them q must be positive; in the latter it is sufficient if $q + 1$ be positive.

99. We have W the same function of $(x, \dots, z, e)$ that V is of $(a, \dots, c, e)$; viz. writing λ for the positive root of

$$\frac{x^2}{f^2 + \lambda} + \dots + \frac{z^2}{h^2 + \lambda} + \frac{e^2}{\lambda} = 1,$$

the value of W is

$$W = \int_{\lambda}^{\infty} t^{q-1} \{(t + f^2) \dots (t + h^2)\}^{-\frac{1}{2}} dt.$$

Considering the formula which involves $e^{2q}W$, — first, if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$, then, when e is $= 0$ the value of λ is not $= 0$; the integral W is therefore finite (not indefinitely large), and we have $e^{2q}W = 0$, consequently $\rho = 0$.

But if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$, then, when e is indefinitely small λ is also indefinitely small; viz. we then have $\frac{e^2}{\lambda} = 1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}$, and the value of W is

$$W = \frac{1}{q} \lambda^q (f \dots h)^{-1},$$

and hence

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{(\sqrt{\pi})^s \Gamma(q + 1)} (f \dots h)^{-1} \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q.$$

[p. 384] **100.** Again, using the formula which involves $(e^{2q+1} \frac{dW}{de})$, we have first $\frac{dV}{da} = -\Theta \frac{d\theta}{da}$, or substituting for Θ and $\frac{d\theta}{da}$ their values and multiplying by e^{2q+1} , we find

$$e^{2q+1} \frac{dV}{da} = 2e^{2q} \theta^{q-1} \frac{a}{(f^2 + \theta)^2} \{(\theta + f^2) \dots (\theta + h^2)\}^{-\frac{1}{2}} \cdot \frac{1}{(\theta + f^2)^2} + \dots + \frac{1}{(h^2 + \theta)^2} + \frac{1}{\theta^2},$$

and therefore

$$e^{2q+1} \frac{dW}{de} = -2e^{2q} \lambda^{q-1} \{(\lambda + f^2) \dots (\lambda + h^2)\}^{-\frac{1}{2}} \frac{1}{\dots}.$$

Hence, writing $e = 0$: first, for an exterior point or $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$, λ is not $= 0$, and the expression vanishes in virtue of the factor e^{2q+1} , whence also $\rho = 0$; next, for an interior point or $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$, λ is $= 0$, hence also $\frac{e^2}{\lambda} = 1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}$ is infinite; neglecting in comparison with it the other terms $\frac{a^2}{(\lambda + f^2)^2} + \dots$, the value is

$$= \frac{2}{e^2} \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{-1};$$

and we have, as before,

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{(\sqrt{\pi})^s \Gamma(q + 1)} (f \dots h)^{-1} \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q.$$

101. Hence in the formula

$$V = \int \frac{\rho dx \dots dz}{\{(a - x)^2 + \dots + (c - z)^2 + e^2\}^{\frac{1}{2}s + q}}$$

ρ has the value just found, or, what is the same thing, we have

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q dx \dots dz}{\{(a - x)^2 + \dots + (c - z)^2 + e^2\}^{\frac{1}{2}s + q}}$$

taken over ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$,

$$= \frac{(\sqrt{\pi})^s \Gamma(q + 1)}{\Gamma(\frac{1}{2}s + q)} (f \dots h) \int_{\theta}^{\infty} t^{q-1} \{(t + f^2) \dots (t + h^2)\}^{-\frac{1}{2}} dt.$$

[p. 385] 102. We may in this result write $e = 0$. There are two cases, according as the attracted point is exterior or interior: if it is exterior, $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$, θ will denote the positive root of the equation $\frac{a^2}{f^2+\theta} + \dots + \frac{c^2}{h^2+\theta} = 1$; if it be interior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} < 1$, θ will be $= 0$; and we thus have

$$\begin{aligned} & \int \frac{(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2})^q dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}} \\ &= \frac{(\sqrt{\pi})^s \Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}} dt, \quad \text{for exterior point } \frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} > 1, \\ &= \frac{(\sqrt{\pi})^s \Gamma(q+1)}{\Gamma(\frac{1}{2}s+q)} (f \dots h) \int_0^{\infty} t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}} dt, \quad \text{for interior point } \frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} < 1; \end{aligned}$$

but as regards the value for an interior point it is to be observed that, unless q be negative (between 0 and -1 , since $1+q$ is positive by hypothesis), the two sides of the equation will be each of them infinite.

103. Third example. We assume here

$$V = \int_{\theta}^{\infty} dt I^m T,$$

where

$$\begin{aligned} I &= 1 - \frac{a^2}{f^2+t} - \dots - \frac{c^2}{h^2+t} - \frac{e^2}{t}, \\ T &= t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}}; \end{aligned}$$

as before, θ is the positive root of the equation

$$1 - \frac{a^2}{f^2+\theta} - \dots - \frac{c^2}{h^2+\theta} - \frac{e^2}{\theta} = 0,$$

and $\frac{1}{2}s+q$ is positive in order that the integral may be finite; also m is positive.

104. In order to show that V satisfies the prepotential equation $\square V = 0$, I shall, in the first place, consider the more general expression

$$V = \int_{\theta+\eta}^{\infty} dt I_0^m T,$$

where η is a constant positive quantity which will be ultimately put $= 0$. The functions previously called J and Θ will be written J_0 and Θ_0 , and J_{η} will now denote

$$J_{\eta} = 1 - \frac{a^2}{\theta+\eta+f^2} - \dots - \frac{c^2}{\theta+\eta+h^2} - \frac{e^2}{\theta+\eta},$$

$$\Theta_{\eta} = (\theta+\eta)^{q-1} \{(\theta+\eta+f^2) \dots (\theta+\eta+h^2)\}^{-\frac{1}{2}};$$

[p. 386] whence also, subtracting from J the evanescent function J_0 , we have

$$J - J_0 = \frac{a^2 \eta}{(\theta+f^2)(\theta+\eta+f^2)} + \dots + \frac{e^2 \eta}{\theta(\theta+\eta)};$$

say this is $= \eta P$; and we have thence, by former equations and in the present notation,

$$\frac{a}{\theta+\eta+f^2} \frac{d\theta}{da} + \dots + \frac{c}{\theta+\eta+h^2} \frac{d\theta}{dc} + \frac{e}{\theta+\eta} \frac{d\theta}{de} = \frac{2}{J'_0} (1 - \eta P),$$

$$\nabla_a \theta = \frac{4}{J_0'^2}, \quad \square \theta = \frac{4\Theta'_0}{\Theta_0 J_0}.$$

In virtue of the equation which determines θ , we have

$$\frac{dV}{da} = \int_{\theta+\eta}^{\infty} dt m I^{m-1} \frac{dI}{da} T, \quad \frac{d^2 V}{da^2} = \int_{\theta+\eta}^{\infty} dt \{m(m-1) I^{m-2} \left(\frac{dI}{da}\right)^2 T + m I^{m-1} \frac{d^2 I}{da^2} T\}$$

$$-J_\eta^m \Theta_\eta \frac{d\theta}{da};$$

and thence

$$\begin{aligned} & -mJ_\eta^{m-1} \Theta_\eta \frac{d\theta}{da} - J_\eta^m \Theta_\eta \frac{d^2\theta}{da^2} - J_\eta^m \Theta_\eta \left(\frac{d\theta}{da} \right)^2 \\ & - J_\eta^m \Theta'_\eta \frac{d\theta}{da}, \end{aligned}$$

with like expressions for $\frac{d^2V}{dc^2}, \frac{d^2V}{de^2}$. Also

$$\frac{2q+1}{e} \frac{dV}{de} = \int_{\theta+\eta}^{\infty} dt mI^{m-1} \frac{2q+1}{e} \frac{dI}{de} T - J_\eta^m \Theta_\eta \cdot \frac{2q+1}{e} \frac{d\theta}{de};$$

and hence

$$\begin{aligned} \square V &= \int_{\theta+\eta}^{\infty} dt \left\{ m(m-1)I^{m-2} \left[\left(\frac{dI}{da} \right)^2 + \dots \right] T + mI^{m-1} \square IT \right\} \\ &+ 4mJ_\eta^{m-1} \Theta_\eta \left(\frac{a}{\theta+\eta+f^2} \frac{d\theta}{da} + \dots + \frac{e}{\theta+\eta} \frac{d\theta}{de} \right) - J_\eta^m \Theta_\eta \square \theta - J_\eta^m \Theta'_\eta \nabla_a \theta. \end{aligned}$$

[p. 387] 105. Writing I', T' for the first derived coefficients of I, T in regard to t , we have

$$I' = \frac{a^2}{(f^2+t)^2} + \dots + \frac{c^2}{(h^2+t)^2} + \frac{e^2}{t^2}, \quad \frac{T'}{T} = \frac{q-1}{t} - \frac{1}{2} \left(\frac{1}{f^2+t} + \dots + \frac{1}{h^2+t} \right) = \frac{2q+1}{2t} \dots$$

The integral is therefore

$$\begin{aligned} & \int_{\theta+\eta}^{\infty} dt \left\{ 2mI^{m-1} I' T + m(m-1)I^{m-2} \cdot 4I' \cdot T \right\} \\ &= \int_{\theta+\eta}^{\infty} dt \left\{ 4mI^{m-1} T' + 4m(m-1)I^{m-2} I' T \right\}, \end{aligned}$$

viz. $I^{m-1}T$ vanishing for $t = \infty$, this is

$$= -4mJ_\eta^{m-1} \Theta.$$

Hence, writing $(J^m \Theta)'$ instead of $\frac{d}{dt}(J^m \Theta)$, we have

$$\begin{aligned} \square V &= -4mJ_\eta^{m-1} \Theta_\eta + 4mJ_\eta^{m-1} \Theta_\eta \left(\frac{a}{\theta+\eta+f^2} \frac{d\theta}{da} + \dots + \frac{e}{\theta+\eta} \frac{d\theta}{de} \right) \\ &- (J^m \Theta)' \nabla_a \theta, \end{aligned}$$

viz. this is

$$\square V = -4mJ_\eta^{m-1} \Theta_\eta + 4mJ_\eta^{m-1} \frac{1}{J'_0} (1 - \eta P) \cdot 2 - (J_\eta^m \Theta_\eta)' \frac{4}{J'^2_0},$$

or, writing $mJ_\eta^{m-1} J'_\eta \Theta + J_\eta^m \Theta'$ instead of $(J^m \Theta)'$, this is

$$\square V = -\frac{4mJ_\eta^{m-1} \Theta_\eta}{J'_0} (J'_\eta - 2JP + J'_0) - \frac{4J_\eta^m}{J'^2_0} (\Theta'_\eta J_0 - \Theta(J'_0)').$$

We have here

$$J'_\eta - 2JP + J'_0 = \eta^2 Q, \quad \text{suppose.}$$

Also $\Theta'_\eta \Theta_0 - \Theta \Theta'_0$ contains the factor η , is $= \eta M$ suppose.

[p. 388] 106. Substituting for $J, J' - 2JP + J$, and $\Theta' \Theta_0 - \Theta \Theta'_0$ their values $\eta P, \eta Q$, and ηM , the whole result contains the factor η^{m+1} , viz. we have

$$\square V = -\frac{4\eta^{m+1} \Theta_\eta}{J_0^{m+1}} (mP^m Q + P^m M).$$

If here, except in the term η^{m+1} , we write $\eta = 0$, we have

$$P = \frac{a^2}{(\theta + f^2)^2} + \cdots + \frac{c^2}{(\theta + h^2)^2} + \frac{e^2}{\theta^2},$$

the formula becomes

$$\square V = -\frac{4\eta^{m+1}\Theta_0}{J_0^{m+1}} \left(mP^m J_0'^- + P^m M_0 \right),$$

or (instead of J_0, Θ_0) using now J, Θ in their original significations

$$J = 1 - \frac{a^2}{f^2 + \theta} - \cdots - \frac{c^2}{h^2 + \theta} - \frac{e^2}{\theta}, \quad \text{and} \quad \Theta = \theta^{q-1} \{(\theta + f^2) \dots (\theta + h^2)\}^{-\frac{1}{2}},$$

this is

$$\square V = -\frac{4\eta^{m+1}\Theta}{J^{m+1}} \left(mP^m J_0^{-1} + P^m M_0 \right),$$

or, what is the same thing,

$$\square V = -\frac{4\eta^{m+1}\Theta}{J^{m+1}} (mP + (\dots)),$$

viz. the expression in $\{ \}$ is

$$-\frac{1}{(\theta + f^2)^2} + \cdots + \frac{1}{(\theta + h^2)^2} + \frac{1}{\theta^2} - \frac{2q+2}{\theta}.$$

We thus see that, η being infinitesimal, $\square V$ is infinitesimal of the order η^{m+1} ; and hence, η being $= 0$, we have

$$\square V = 0;$$

viz. the prepotential equation is satisfied by the value

$$V = \int_{\theta}^{\infty} dt I^m T,$$

where $m+1$ is positive.

107. We have consequently a value of ρ corresponding to the foregoing value of V ; and this value is

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q+1)} \frac{1}{e^{2q+1}} \frac{dW}{de},$$

[p. 389] where, writing λ for the positive root of

$$1 - \frac{x^2}{f^2 + \lambda} - \cdots - \frac{z^2}{h^2 + \lambda} - \frac{e^2}{\lambda} = 0,$$

we have

$$W = \int_{\lambda}^{\infty} dt \left(1 - \frac{x^2}{f^2 + t} - \cdots - \frac{z^2}{h^2 + t} - \frac{e^2}{t} \right)^m t^{q-1} \{ (t + f^2) \dots (t + h^2) \}^{-\frac{1}{2}};$$

we thence obtain

$$\frac{dW}{de} = \int_{\lambda}^{\infty} dt \frac{-2me}{t} \left(1 - \frac{x^2}{f^2 + t} - \cdots - \frac{z^2}{h^2 + t} - \frac{e^2}{t} \right)^{m-1} t^{q-1} \{ (t + f^2) \dots (t + h^2) \}^{-\frac{1}{2}} - (\dots) \frac{d\lambda}{de},$$

or, multiplying by e^{2q+1} and substituting for $\frac{d\lambda}{de}$ its value $\frac{2e}{\lambda} \cdot \frac{1}{\dots}$, we have

$$e^{2q+1} \frac{dW}{de} = -2me^{2q+2} \int_{\lambda}^{\infty} dt \frac{1}{t} \left(1 - \frac{x^2}{f^2 + t} - \cdots - \frac{z^2}{h^2 + t} - \frac{e^2}{t} \right)^{m-1} t^{q-1} \{ (t + f^2) \dots (t + h^2) \}^{-\frac{1}{2}} + \dots,$$

where the second term, although containing the evanescent factor

$$\left(1 - \frac{x^2}{f^2 + \lambda} - \cdots - \frac{z^2}{h^2 + \lambda} - \frac{e^2}{\lambda} \right)^m,$$

is for the present retained.

108. I attend to the second term.

1°. Suppose $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$; then, as e diminishes and becomes $= 0$, λ does not become zero, but it becomes the positive root of the equation

$$1 - \frac{x^2}{f^2 + \lambda} - \dots - \frac{z^2}{h^2 + \lambda} = 0;$$

hence the term, containing as well the evanescent factor e^{2q+2} as the other evanescent factor $\left(1 - \frac{x^2}{f^2 + \lambda} - \dots - \frac{z^2}{h^2 + \lambda} - \frac{e^2}{\lambda}\right)^m$, is $= 0$.

[p. 390] 2°. Suppose $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$; then, as e diminishes to zero, λ tends to become $= 0$, but $\frac{e^2}{\lambda}$ is finite and $= 1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}$, whence $\frac{1}{\lambda}$ is indefinitely large; and since $\frac{x^2}{(\lambda + f^2)^2} + \dots + \frac{z^2}{(\lambda + h^2)^2}$ becomes $= +\frac{x^2}{f^4} + \dots + \frac{z^2}{h^4}$, which is finite, the denominator may be reduced to $\frac{e^2}{\lambda}$, and the term therefore is

$$= -2 \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right) \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{m-1} e^{2q+2} (f \dots h)^{-1} \lambda^{-q-1},$$

which, the other factor being finite, vanishes in virtue of the evanescent factor

$$\left(1 - \frac{x^2}{f^2 + \lambda} - \dots - \frac{z^2}{h^2 + \lambda} - \frac{e^2}{\lambda}\right)^m.$$

Hence the second term always vanishes, and we have (e being $= 0$)

$$e^{2q+1} \frac{dW}{de} = -2m \int_{\lambda}^{\infty} dt \frac{e^{2q+2}}{t} \left(1 - \frac{x^2}{f^2 + t} - \dots - \frac{z^2}{h^2 + t} - \frac{e^2}{t}\right)^{m-1} t^{q-1} \{(t + f^2) \dots (t + h^2)\}^{-\frac{1}{2}}.$$

109. Considering first the case $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$: then, as e diminishes to zero, λ does not become $= 0$; the integral contains no infinite element, and it consequently vanishes in virtue of the factor e^{2q+2} .

But if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$, then, introducing instead of t the new variable ξ , $= t \cdot \frac{1}{\lambda}$, that is, $t = \frac{\lambda}{\xi}$, $dt = -\frac{\lambda}{\xi^2} d\xi$, and writing for shortness

$$R = 1 - \frac{x^2}{f^2 + \xi} - \dots - \frac{z^2}{h^2 + \xi},$$

the term becomes

$$= \int_0^R d\xi \cdot 2m(R - \xi)^{m-1} \xi^q \{(f^2 + \xi) \dots (h^2 + \xi)\}^{-\frac{1}{2}},$$

where, as regards the limits, corresponding to $t = \infty$ we have $\xi = 0$, and corresponding to $t = \lambda$ we have ξ the positive root of $R - \xi = 0$. But e is indefinitely small; except for indefinitely small values of ξ , we have

$$R = 1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}, \quad \text{and} \quad \{(f^2 + \xi) \dots (h^2 + \xi)\}^{-\frac{1}{2}} = (f \dots h)^{-1};$$

[p. 391] and if ξ be indefinitely small, then, whether we take the accurate or the reduced expressions, the elements are finite, and the corresponding portion of the integral is indefinitely small. We may consequently reduce as above; viz. writing now

$$R = 1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2},$$

the formula is

$$\begin{aligned} e^{2q+1} \frac{dW}{de} &= \int_0^R d\xi \, 2m(R - \xi)^{m-1} \xi^q (f \dots h)^{-1} \\ &= -2m(f \dots h)^{-1} \int_0^R d\xi \, \xi^q (R - \xi)^{m-1}, \end{aligned}$$

or writing $\xi = Ru$, the integral becomes $= R^{q+m} \int_0^1 du u^q (1-u)^{m-1}$, which is

$$= R^{q+m} \frac{\Gamma(1+q)\Gamma(m)}{\Gamma(1+q+m)};$$

that is, we have

$$e^{2q+1} \frac{dW}{de} = - \frac{2m(f \dots h)^{-1} \Gamma(1+q)\Gamma(m)}{\Gamma(1+q+m)} R^{q+m},$$

and consequently

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)\Gamma(m)}{(\sqrt{\pi})^s \Gamma(1+q+m)} (f \dots h)^{-1} R^{q+m},$$

that is,

$$\rho = \frac{\Gamma(\frac{1}{2}s+q)\Gamma(m)}{(\sqrt{\pi})^s \Gamma(1+q+m)} (f \dots h)^{-1} \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{q+m},$$

viz. ρ has this value for values of $(x, \dots, z)$ such that $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$, but is $= 0$ if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$.

110. Multiplying by a constant factor so as to reduce ρ to the value R^{q+m} , the final result is that the integral

$$V = \int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{q+m} dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}},$$

the limits being given by the equation

$$\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1,$$

is equal to

$$\frac{\Gamma(\frac{1}{2}s)\Gamma(1+q+m)}{\Gamma(1+q)\Gamma(\frac{1}{2}s+q+m)} (f \dots h) \int_{\theta}^{\infty} dt t^m \left(1 - \frac{a^2}{f^2+t} - \dots - \frac{c^2}{h^2+t} - \frac{e^2}{t}\right)^m t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}},$$

where θ is the positive root of

$$1 - \frac{a^2}{\theta+f^2} - \dots - \frac{c^2}{\theta+h^2} - \frac{e^2}{\theta} = 0.$$

[p. 392] In particular, if $e = 0$, or

$$V = \int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{q+m} dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2\}^{\frac{1}{2}s+q}},$$

there are two cases:

exterior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} > 1$, θ is positive root of $1 - \frac{a^2}{f^2+\theta} - \dots - \frac{c^2}{h^2+\theta} = 0$,

interior, $\frac{a^2}{f^2} + \dots + \frac{c^2}{h^2} < 1$, θ vanishes, viz. the limits in the integral are $\infty, 0$;

q must be negative, $1+q$ positive as before, in order that the t -integral may not be infinite in regard to the element $t = 0$.

It is assumed in the proof that m and $1+q$ are each of them positive; but, as appears by the second example, the theorem is true for the extreme value $m = 0$; it does not, however, appear that the proof can be extended to include the extreme value $q = -1$. The formula seems, however, to hold good for values of m, q beyond the foregoing limits; and it would seem that the only necessary conditions are $\frac{1}{2}s+q$, $1+m$, and $1+q+m$, each of them positive. The theorem is, in fact, a particular case of the following one, proved Annex X. No. 162, viz.

$$V = \int \frac{\phi\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right) dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}},$$

taken over the ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$, is equal to

$$\frac{(\sqrt{\pi})^s \Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} dt \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}} (1-\sigma)^{q+\frac{1}{2}s} \int_0^1 (1-x)^q \phi\{\sigma + (1-\sigma)x\} dx,$$

where σ denotes $\frac{a^2}{f^2+t} + \dots + \frac{c^2}{h^2+t} + \frac{e^2}{t}$; assuming $\phi u = (1-u)^{q+m}$, we have

$$\phi\{\sigma + (1-\sigma)x\} = (1-\sigma)^{q+m} (1-x)^{q+m},$$

and the theorem is thus proved.

111. Particular cases: $m = 0$;

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}} = \frac{(\sqrt{\pi})^s \Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} dt t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}}.$$

Cor. In a somewhat similar manner it may be shown that

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q x dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}} = \frac{a(\sqrt{\pi})^s \Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} dt t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}}.$$

[p. 393] Multiplying the first by a and subtracting it from the second, we have

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q (a-x) dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}} = -\frac{(\sqrt{\pi})^s \Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} dt \frac{a}{f^2+t} t^{q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}},$$

or, writing $q+1$ for q , this is

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q (a-x) dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q+1}} = \frac{(\sqrt{\pi})^s \Gamma(1+q)}{\Gamma(1+\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} dt \frac{a}{f^2+t} t^q \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}},$$

and we have similar formulae with $\dots, (c-z), e$, instead of $(a-x)$, in the numerator.

112. If $m = 1$, we have

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{q+1} dx \dots dz}{\{(a-x)^2 + \dots + (c-z)^2 + e^2\}^{\frac{1}{2}s+q}} = \frac{(\sqrt{\pi})^s \Gamma(1+q)}{\Gamma(2+\frac{1}{2}s+q)} (f \dots h) \int_{\theta}^{\infty} dt t^q \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}},$$

which, differentiated in respect to a , gives the $(a-x)$ -formula; hence conversely, assuming the $a-x, \dots, c-z, e$ -formulae, we obtain by integration the last preceding formula to a constant près, viz. we thereby obtain the multiple integral = C + right-hand function, where C is independent of $(a, \dots, c, e)$; by taking these all infinite, and observing that then $\theta = \infty$, the two integrals each vanish, and we obtain $C = 0$.

In particular, when $s = 3, q = -\frac{1}{2}$, then

$$\int \frac{\left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^{\frac{1}{2}} dx dy dz}{\{(a-x)^2 + (b-y)^2 + (c-z)^2 + e^2\}^1} = \frac{\pi^2}{2} (fgh) \int_{\theta}^{\infty} \frac{dt}{\sqrt{t(t+f^2)(t+g^2)(t+h^2)}},$$

which, putting therein $e = 0$, gives the potential of an ellipsoid for the cases of an exterior point and an interior point respectively.

Annex V. Green's Integration of the Prepotential Equation. Art. Nos. 113 to 128.

[p. 393, cont.] 113. In the present Annex, I in part reproduce Green's process for the integration of this equation by means of a series of functions, which are analogous to Laplace's Functions and may be termed "Greenians" (see his Memoir on the Attraction of Ellipsoids, referred to above, p. 320); each such function gives rise to a Prepotential Integral.

Green shows, by a complicated and difficult piece of general reasoning, that there exist solutions of the form $V = \Theta\phi$ (see post. No. 116), where ϕ is a function of the

[p. 394] s new variables $\alpha, \beta, \dots, \gamma$ without θ , such that $\nabla^2\phi = \kappa\phi$, κ being a function of θ only; these functions ϕ of the variables $\alpha, \beta, \dots, \gamma$ are in fact the Greenian Functions in question. The function of the order 0 is $\phi = 1$; those of the order 1 are $\phi = \alpha, \phi = \beta, \dots, \phi = \gamma$; those of the order 2 are $\phi = \alpha\beta$, etc., and s functions each of the form

$$\phi = \frac{1}{2}(A\alpha^2 + B\beta^2 + \dots + C\gamma^2) + D.$$

The existence of the functions just referred to other than the s functions involving the squares of the variables is obvious enough; the difficulty first arises in regard to these s functions; and the actual development of them appears to me important by reason of the light which is thereby thrown upon the general theory. This I accomplish in the present Annex; and I determine by Green's process the corresponding prepotential integrals. I do not go into the question of the Greenian Functions of orders superior to the second.

114. I write for greater clearness $(a, b, \dots, c, e)$ instead of $(a, \dots, c, e)$ to denote the series of $(s+1)$ variables; viz. $(a, b, \dots, c)$ will denote a series of s variables; corresponding to these we have the semiaxes $(f, g, \dots, h)$, and the new variables $(\alpha, \beta, \dots, \gamma)$; these last, with the before-mentioned function θ , are the $s+1$ new variables of the problem; and, for convenience, there is introduced also a quantity ε ; viz. we have

$$a = \sqrt{f^2 + \theta}\alpha, \quad b = \sqrt{g^2 + \theta}\beta, \quad \dots, \quad c = \sqrt{h^2 + \theta}\gamma, \quad e = \sqrt{\theta}\varepsilon,$$

where $\lambda = \alpha^2 + \beta^2 + \dots + \gamma^2 + \varepsilon^2$.

That is, we have θ a function of $a, b, \dots, c, e$, determined by

$$\frac{a^2}{f^2 + \theta} + \frac{b^2}{g^2 + \theta} + \dots + \frac{c^2}{h^2 + \theta} + \frac{e^2}{\theta} = 1,$$

and then $\alpha, \beta, \dots, \gamma$ are given as functions of the same quantities $a, b, \dots, c, e$ by the equations

$$\alpha^2 = \frac{a^2}{f^2 + \theta}, \quad \beta^2 = \frac{b^2}{g^2 + \theta}, \quad \dots, \quad \gamma^2 = \frac{c^2}{h^2 + \theta};$$

also ε , considered as a function of the same quantities, is

$$\varepsilon^2 = 1 - \frac{a^2}{f^2 + \theta} - \frac{b^2}{g^2 + \theta} - \dots - \frac{c^2}{h^2 + \theta} = \frac{e^2}{\theta}.$$

115. Introducing instead of $a, b, \dots, c, e$ the new variables $\alpha, \beta, \dots, \gamma, \theta$, the transformed differential equation is

$$\nabla^2 V + \frac{d^2 V}{d\theta^2} \left(s + 2q + 2 - \frac{\alpha^2}{f^2 + \theta} - \dots - \frac{\gamma^2}{h^2 + \theta} \right) \cdot 4 + \nabla^2 V = 0,$$

[p. 395] where for shortness

$$\begin{aligned} \nabla^2 V = & \theta \left[\frac{f^2 + \theta}{a^2} - \frac{h^2 + \theta}{c^2} + 1 \right] \frac{d^2 V}{d\alpha^2} + \dots \\ & + \theta \left[\frac{f^2 + \theta}{a^2} + \dots \right] \frac{d^2 V}{d\gamma^2} - \frac{\theta \cdot 2q}{e^2} \frac{d^2 V}{d\varepsilon^2} \dots \\ & + \frac{2\alpha}{f^2 + \theta} \frac{d^2 V}{d\alpha d\beta} - \&c. \\ & + \frac{1}{f^2 + \theta} \left(-\frac{\alpha}{f^2 + \theta} - \&c. \right) \frac{dV}{d\alpha} \\ & + \frac{1}{\theta} \left(-\frac{\varepsilon}{\theta} - \&c. \right) \frac{dV}{d\beta} \end{aligned}$$

$$+\frac{1}{\theta}\left(-\frac{\varepsilon}{\theta}\right)\frac{dV}{d\gamma}.$$

Also

$$\frac{d^2V}{d\theta^2} = \frac{1}{4(f^2 + \theta)} \left[\frac{f^2 + \theta}{a^2} + \cdots + \frac{h^2 + \theta}{c^2} + \frac{\theta}{e^2} \right] \left[(f^2 + \theta) \frac{d^2V}{d\alpha^2} + \cdots + (h^2 + \theta) \frac{d^2V}{d\gamma^2} + \theta \frac{d^2V}{d\varepsilon^2} \right] \cdots.$$

116. To integrate the equation for V , we assume

$$V = \Theta\phi,$$

where Θ is a function of θ only, and ϕ a function of $\alpha, \beta, \dots, \gamma$ (without θ), such that

$$\nabla^2\phi = \kappa\phi,$$

κ being a function of θ only. Assuming that this is possible, the remaining equation to be satisfied is obviously

$$\Theta'' + \kappa\Theta = 0.$$

Solutions of the form in question are

$$\begin{aligned} \phi &= 1, \quad \kappa = 0, \\ \phi &= \alpha, \quad \kappa = \frac{1}{f^2 + \theta}, \quad \text{or} \quad \phi = \beta, \dots, \kappa = \frac{1}{g^2 + \theta}, \dots, \frac{1}{h^2 + \theta}, \\ \phi &= \alpha\beta, \quad \kappa = \frac{1}{f^2 + \theta} + \frac{1}{g^2 + \theta}, \quad \&c., \end{aligned}$$

and the corresponding values of Θ involve θ .

[p. 396] and it can be shown next that there is a solution of the form

$$\phi = \frac{1}{2}(A\alpha^2 + \cdots + C\gamma^2) + D.$$

117. In fact, assuming that this satisfies $\nabla^2\phi - \kappa\phi = 0$, we must have identically

$$\begin{aligned} &\frac{A}{f^2 + \theta} \left\{ -\alpha^2 \frac{\theta}{f^2 + \theta} (\beta^2 + \cdots + \gamma^2) - \alpha^2 \frac{\theta}{f^2 + \theta} \varepsilon^2 + 1 \right\} \\ &\quad + \frac{B}{g^2 + \theta} \left\{ \cdots \right\} + \cdots \\ &\quad + \frac{2q}{\varepsilon^2} \left\{ -\alpha^2 \frac{\theta}{f^2 + \theta} - \beta^2 \frac{\theta}{g^2 + \theta} - \cdots \right\} \\ &\quad + \frac{A}{f^2 + \theta} \left\{ -\alpha^2 \frac{\theta}{f^2 + \theta} - \cdots + \frac{2}{f^2 + \theta} \right\} + \cdots \\ &\quad + \frac{2q}{\varepsilon^2} \left\{ -\alpha^2 \frac{\theta}{f^2 + \theta} - \cdots \right\} \\ &\quad + \kappa \left\{ \frac{1}{2}(A\alpha^2 + B\beta^2 + \cdots + C\gamma^2) + D \right\} = 0; \end{aligned}$$

so that, from the term in α^2 , we have

$$\frac{A}{f^2 + \theta} \left\{ -\frac{\theta}{f^2 + \theta} (s + 2q + 2) \right\} - \frac{B\theta}{(g^2 + \theta)(f^2 + \theta)} - \cdots - \frac{C\theta}{(h^2 + \theta)(f^2 + \theta)} + \frac{1}{2}\kappa A = 0,$$

or, what is the same thing,

$$A \left\{ 2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta) \right\} + \kappa f^2 D = 0,$$

with the like equations from $\beta^2, \dots, \gamma^2$; and from the constant term we have

$$\frac{A}{f^2 + \theta} + \frac{B}{g^2 + \theta} + \cdots + \frac{C}{h^2 + \theta} + \kappa D = 0.$$

118. Multiplying this last by f^2 , and adding it to the first, we obtain

$$A\left\{1 + \frac{f^2}{f^2 + \theta}\right\} \cdot 1 + \frac{Bf^2}{g^2 + \theta} + \cdots + \frac{Cf^2}{h^2 + \theta} + \frac{1}{2}\kappa A(f^2 + \theta) + \kappa f^2 D = 0;$$

viz. putting for shortness $\Omega = \frac{a^2}{f^2 + \theta} + \cdots = \frac{\theta}{f^2 + \theta}\alpha^2 + \cdots + \frac{\theta}{h^2 + \theta}\gamma^2$, this is

$$A\{2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta)\} + \kappa f^2 D = 0;$$

and similarly

$$B\{2q + 2 + \Omega + \frac{1}{2}\kappa(g^2 + \theta)\} + \kappa g^2 D = 0,$$

$$C\{2q + 2 + \Omega + \frac{1}{2}\kappa(h^2 + \theta)\} + \kappa h^2 D = 0.$$

[p. 397] To these we join the foregoing equation

$$\frac{A}{f^2 + \theta} + \frac{B}{g^2 + \theta} + \cdots + \frac{C}{h^2 + \theta} + \kappa D = 0.$$

Eliminating $A, B, \dots, C, D$, we have an equation which determines κ as a function of θ ; and the equations then determine the ratios of $A, B, \dots, C, D$, so that these quantities will be given as determinate multiples of an arbitrary quantity M . The equation for κ is in fact

$$\frac{f^2(f^2 + \theta)}{2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta)} + \cdots + \frac{h^2(h^2 + \theta)}{2q + 2 + \Omega + \frac{1}{2}\kappa(h^2 + \theta)} + 1 = 0;$$

and the values of $A, B, \dots, C, D$ are then

$$A = \frac{Mf^2}{2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta)}, \quad B = \frac{Mg^2}{2q + 2 + \Omega + \frac{1}{2}\kappa(g^2 + \theta)}, \quad \dots, \quad C = \frac{Mh^2}{2q + 2 + \Omega + \frac{1}{2}\kappa(h^2 + \theta)}, \quad D = -\frac{M}{\kappa},$$

values which seem to be dependent on θ : if they were so, it would be fatal to the success of the process; but they are really independent of θ .

119. That they are independent of θ depends on the theorems; that we have

$$\kappa = \frac{(2q + 2 + \Omega)\kappa_0}{2q + 2 - \frac{1}{2}\kappa_0\theta},$$

where κ_0 is a quantity independent of θ determined by the equation

$$\frac{f^2}{2q + 2 + \frac{1}{2}\kappa_0 f^2} + \frac{g^2}{2q + 2 + \frac{1}{2}\kappa_0 g^2} + \cdots + \frac{h^2}{2q + 2 + \frac{1}{2}\kappa_0 h^2} + 1 = 0$$

(κ_0 is in fact the value of κ on writing $\theta = 0$): and that, omitting the arbitrary multiplier, the values of $A, B, \dots, C, D$ then are

$$\frac{f^2}{2q + 2 + \frac{1}{2}\kappa_0 f^2}, \quad \frac{g^2}{2q + 2 + \frac{1}{2}\kappa_0 g^2}, \quad \dots, \quad \frac{h^2}{2q + 2 + \frac{1}{2}\kappa_0 h^2}, \quad -\frac{1}{\kappa_0},$$

or, what is the same thing, the value of ϕ is

$$\phi = \frac{f^2\alpha^2}{2q + 2 + \frac{1}{2}\kappa_0 f^2} + \frac{g^2\beta^2}{2q + 2 + \frac{1}{2}\kappa_0 g^2} + \cdots + \frac{h^2\gamma^2}{2q + 2 + \frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}.$$

120. To explain the ground of the assumption

$$\kappa = \frac{(2q + 2 + \Omega)\kappa_0}{2q + 2 - \frac{1}{2}\kappa_0\theta},$$

observe that, assuming

$$\frac{2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta)}{2q + 2 + \frac{1}{2}\kappa_0 f^2} = \frac{2q + 2 + \Omega + \frac{1}{2}\kappa(g^2 + \theta)}{2q + 2 + \frac{1}{2}\kappa_0 g^2},$$

[p. 398] then multiplying out and reducing, we obtain

$$(g^2 - f^2) \{ \kappa_0(2q + 2 + \Omega) - (2q + 2)\kappa + \frac{1}{2}\kappa\kappa_0\theta \} = 0;$$

viz. the equation divides out by the factor $g^2 - f^2$, thereby becoming

$$\kappa_0(2q + 2 + \Omega) - (2q + 2)\kappa + \frac{1}{2}\kappa\kappa_0\theta = 0,$$

that is, it gives for κ the foregoing value: hence clearly, κ having this value, we obtain by symmetry

$$2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta), \quad 2q + 2 + \Omega + \frac{1}{2}\kappa(g^2 + \theta), \quad \dots, \quad 2q + 2 + \Omega + \frac{1}{2}\kappa(h^2 + \theta),$$

proportional to

$$2q + 2 + \frac{1}{2}\kappa_0 f^2, \quad 2q + 2 + \frac{1}{2}\kappa_0 g^2, \quad \dots, \quad 2q + 2 + \frac{1}{2}\kappa_0 h^2;$$

viz. the ratios, not only of $A : B$, but of $A : B : \dots : C$ will be independent of θ .

121. To complete the transformation, starting with the foregoing value of κ , we have

$$2q + 2 + \Omega + \frac{1}{2}\kappa(f^2 + \theta) = (2q + 2 + \Omega) \frac{2q + 2 + \frac{1}{2}\kappa_0(f^2 + \theta)}{2q + 2 - \frac{1}{2}\kappa_0\theta}, \quad \&c.;$$

so that we have

$$\begin{aligned} A\{2q + 2 + \frac{1}{2}\kappa_0 f^2\} + \kappa_0 f^2 D &= 0, \\ B\{2q + 2 + \frac{1}{2}\kappa_0 g^2\} + \kappa_0 g^2 D &= 0, \\ C\{2q + 2 + \frac{1}{2}\kappa_0 h^2\} + \kappa_0 h^2 D &= 0, \end{aligned}$$

and

$$\frac{A}{f^2 + \theta} + \frac{B}{g^2 + \theta} + \dots + \frac{C}{h^2 + \theta} = \frac{(2q + 2 + \Omega)\kappa_0 D}{2q + 2 - \frac{1}{2}\kappa_0\theta}.$$

Substituting for $A, B, \dots, C$ their values, this last becomes

$$\frac{\kappa_0 f^2 D}{2q + 2 + \frac{1}{2}\kappa_0\theta} \left\{ \frac{2q + 2}{2q + 2 + \frac{1}{2}\kappa_0 f^2} - \dots - \frac{\kappa_0 h^2 D}{h^2 + \theta} \right\} + \frac{2q + 2}{2q + 2 - \frac{1}{2}\kappa_0\theta} \left\{ \frac{2q + 2}{2q + 2 + \frac{1}{2}\kappa_0 h^2} - \frac{\theta}{h^2 + \theta} \right\};$$

viz. this is

$$\left\{ \frac{2q + 2}{2q + 2 + \frac{1}{2}\kappa_0 f^2} - \dots \right\} (2q + 2)\kappa_0 D,$$

or, substituting for κ its value, and dividing out by $2q + 2$, we have

$$\frac{1}{2q + 2 + \frac{1}{2}\kappa_0 f^2} + \frac{1}{2q + 2 + \frac{1}{2}\kappa_0 g^2} + \dots + \frac{1}{2q + 2 + \frac{1}{2}\kappa_0 h^2} = -1,$$

the equation for the determination of κ_0 .

[p. 399] **122.** The equation for κ_0 is of the order s ; there are consequently s functions of the form in question, and each of the terms $\alpha^2, \beta^2, \dots, \gamma^2$ can be expressed as a linear function of these. It thus appears that any quadric function of $\alpha, \beta, \dots, \gamma$ can be expressed as a sum of Greenian functions; viz. the form is

$$\begin{aligned} &A\alpha^2 + B\beta^2 + \&c. \\ &+ C'\alpha\beta + \&c. \\ &+ D' \left(\frac{f^2 \alpha^2}{2q + 2 + \frac{1}{2}\kappa'_0 f^2} + \frac{g^2 \beta^2}{2q + 2 + \frac{1}{2}\kappa'_0 g^2} + \dots + \frac{h^2 \gamma^2}{2q + 2 + \frac{1}{2}\kappa'_0 h^2} - \frac{1}{\kappa'_0} \right) \\ &+ D'' (\quad , \quad \dots, \quad \dots, \quad \dots) \\ &(s \text{ lines}), \end{aligned}$$

viz. the terms multiplied by $D', D'', \&c.$ respectively are those answering to the roots $\kappa'_0, \kappa''_0, \dots$ of the equation in κ_0 .

The general conclusion is that any rational and integral function of $\alpha, \beta, \dots, \gamma$ can be expressed as a sum of Greenian functions.

123. We have next to integrate the equation

$$\Theta'' + \kappa\Theta = 0.$$

Suppose $\kappa = 0$, a particular solution is $\Theta = 1$. Next, suppose

$$\kappa = \frac{1}{f^2 + \theta} \cdot \frac{-1}{2q-2} \cdot \frac{-\theta}{f^2 + \theta} - \cdots - \frac{-\theta}{h^2 + \theta} - 1;$$

a particular solution is $\Theta = \sqrt{f^2 + \theta}$: in fact, omitting the constant denominator, or writing $\Theta = \sqrt{f^2 + \theta}$, and therefore

$$\frac{d\Theta}{d\theta} = \frac{1}{2} \frac{1}{\sqrt{f^2 + \theta}}, \quad \frac{d^2\Theta}{d\theta^2} = -\frac{1}{4} \frac{1}{(f^2 + \theta)^{\frac{3}{2}}},$$

the equation to be verified is

$$-\frac{1}{4} \frac{1}{(f^2 + \theta)^{\frac{3}{2}}} + \frac{1}{f^2 + \theta} \cdot \sqrt{f^2 + \theta} \cdot \left\{ 2q - 2 - \frac{-\theta}{f^2 + \theta} - \cdots \right\} = 0.$$

Again, suppose $\kappa = \frac{1}{f^2 + \theta} \cdot \frac{1}{g^2 + \theta} + \frac{2\theta}{(f^2 + \theta)(g^2 + \theta)} + \frac{1}{f^2 + \theta} \cdot \frac{1}{g^2 + \theta}$ &c. (value belonging to $\phi = \alpha\beta$, see No. 116);

a particular solution is $\Theta = \sqrt{f^2 + \theta} \cdot \sqrt{g^2 + \theta}$: in fact, omitting the constant factor, or writing
[p. 400] and therefore

$$\frac{d\Theta}{d\theta} = \sqrt{\frac{g^2 + \theta}{f^2 + \theta}} \cdot \frac{1}{2} + \sqrt{\frac{f^2 + \theta}{g^2 + \theta}} \cdot \frac{1}{2}, \quad \frac{d^2\Theta}{d\theta^2} = -\frac{1}{4} \frac{\sqrt{f^2 + \theta}}{(g^2 + \theta)^{\frac{3}{2}}} - \frac{1}{4} \frac{\sqrt{g^2 + \theta}}{(f^2 + \theta)^{\frac{3}{2}}},$$

the equation to be verified is

$$-\frac{1}{4} \left\{ \frac{\sqrt{g^2 + \theta}}{(f^2 + \theta)^{\frac{3}{2}}} + \frac{2\theta}{\sqrt{f^2 + \theta} \sqrt{g^2 + \theta}} + \frac{\sqrt{f^2 + \theta}}{(g^2 + \theta)^{\frac{3}{2}}} \right\} \\ + (\sqrt{g^2 + \theta} + \sqrt{f^2 + \theta}) \cdot \frac{1}{4} = 0,$$

or putting for shortness $\Omega = \frac{\theta}{f^2 + \theta} + \frac{\theta}{g^2 + \theta} + \cdots + \frac{\theta}{h^2 + \theta}$, this is

$$\frac{\theta}{(f^2 + \theta)^2} \cdot \sqrt{g^2 + \theta} - \frac{2\theta}{\sqrt{f^2 + \theta} \sqrt{g^2 + \theta}} + \frac{\theta}{(g^2 + \theta)^2} \cdot \sqrt{f^2 + \theta} + \frac{2\theta}{\sqrt{f^2 + \theta} \sqrt{g^2 + \theta}} \{2q+2+\Omega\} \cdot (-\sqrt{g^2 + \theta} - \sqrt{f^2 + \theta}) =$$

which is true. And, generally, the particular solution is deduced from the value of ϕ by writing therein

$$\frac{\sqrt{f^2 + \theta}}{\sqrt{f^2 + g^2 + \cdots + h^2}}, \quad \frac{\sqrt{g^2 + \theta}}{\sqrt{f^2 + g^2 + \cdots + h^2}}, \quad \cdots, \quad \frac{\sqrt{h^2 + \theta}}{\sqrt{f^2 + g^2 + \cdots + h^2}}$$

in place of $\alpha, \beta, \dots, \gamma$ respectively: say the value thus obtained is $\Theta = H$, where H is what ϕ becomes by the above substitution.

124. Represent for a moment the equation in Θ by

$$\frac{d^2\Theta}{d\theta^2} + 2P \frac{d\Theta}{d\theta} + \kappa\Theta = 0,$$

and assume that this is satisfied by $\Theta = H \int z d\theta$. Then we have

$$\frac{d\Theta}{d\theta} = zH + \frac{dH}{d\theta} \int z d\theta,$$

$$\frac{d^2\Theta}{d\theta^2} = z \frac{dH}{d\theta} + H \frac{dz}{d\theta} + \frac{d^2H}{d\theta^2} \int z d\theta,$$

$$+\kappa H \int z d\theta = 0,$$

[p. 401] and therefore

$$(H'' + 2PH' + \kappa H) \int z d\theta + 2Hz' + 4PHz = 0;$$

viz. multiplying by $\frac{H}{4z}$, this is

$$\frac{H^2 z'}{2z} + 2PH^2 = 0,$$

or

$$\frac{d}{d\theta}(H^2 z) + 4HH'z = 0;$$

viz. substituting for P its value, this is

$$\frac{1}{H^2 z} \frac{d}{d\theta}(H^2 z) \left(\frac{1}{2}q + 1 - \frac{1}{f^2 + \theta} - \cdots - \frac{1}{h^2 + \theta} \right) \cdot \frac{1}{2} = 0.$$

Hence, integrating,

$$H^2 z = \frac{C\theta^{-q-1}}{\sqrt{f^2 + \theta} \cdot \sqrt{g^2 + \theta} \cdots \sqrt{h^2 + \theta}}, \quad C \text{ an arbitrary constant,}$$

and

$$\Theta = CH \int_{\chi}^{\theta} \frac{\theta^{-q-1} d\theta}{H^2 \sqrt{f^2 + \theta} \cdot \sqrt{g^2 + \theta} \cdots \sqrt{h^2 + \theta}}, \quad \chi \text{ arbitrary,}$$

where the constants of integration are C, χ ; or, what is the same thing, taking $\bar{T}$ the same function of t that Θ is of θ (viz. $\bar{T}$ is what H becomes on writing therein

$$\frac{\sqrt{f^2 + t}}{\sqrt{f^2 + g^2 + \cdots + h^2}}, \quad \frac{\sqrt{g^2 + t}}{\sqrt{f^2 + g^2 + \cdots + h^2}}, \quad \cdots, \quad \frac{\sqrt{h^2 + t}}{\sqrt{f^2 + g^2 + \cdots + h^2}}$$

in place of $\alpha, \beta, \dots, \gamma$ respectively), then

$$\Theta = -CH \int_{\theta}^{\chi} \frac{t^{-q-1} dt}{T^2 \sqrt{f^2 + t} \cdot \sqrt{g^2 + t} \cdots \sqrt{h^2 + t}},$$

where χ may be taken $= \infty$: we thus have

$$V = \Theta\phi = -CH\phi \int_{\theta}^{\infty} \frac{t^{-q-1} dt}{T^2 \sqrt{f^2 + t} \cdot \sqrt{g^2 + t} \cdots \sqrt{h^2 + t}}.$$

Recollecting that

$$1 = \frac{a^2}{f^2 + \theta} + \frac{b^2}{g^2 + \theta} + \cdots + \frac{c^2}{h^2 + \theta} + \frac{e^2}{\theta},$$

so that for $\theta = \infty$ we have $a^2 + b^2 + \cdots + c^2 + e^2 = \theta$, the assumption $\chi = \infty$ comes to making V vanish for infinite values of $(a, b, \dots, c, e)$.

125. We have to find the value of ρ corresponding to the foregoing value of V ; viz. W being the value of V , on writing therein $(x, y, \dots, z)$ in place of $(a, b, \dots, c)$, then (theorem A)

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q + 1) e^{2q+1}} \cdot \frac{dW}{de}.$$

[p. 402] Take λ the same function of $(x, y, \dots, z, e)$ that θ is of $(a, b, \dots, c, e)$: viz. take λ the positive root of

$$\frac{x^2}{f^2 + \lambda} + \frac{y^2}{g^2 + \lambda} + \cdots + \frac{z^2}{h^2 + \lambda} + \frac{e^2}{\lambda} = 1;$$

and let $(\xi, \eta, \dots, \zeta, \iota)$ correspond to $(\alpha, \beta, \dots, \gamma, \varepsilon)$, viz.

$$\xi = \frac{x}{\sqrt{f^2 + \lambda}}, \quad \eta = \frac{y}{\sqrt{g^2 + \lambda}}, \quad \dots, \quad \zeta = \frac{z}{\sqrt{h^2 + \lambda}}, \quad \iota = \sqrt{1 - \frac{x^2}{f^2 + \lambda} - \dots - \frac{z^2}{h^2 + \lambda}} = \frac{e}{\sqrt{\lambda}};$$

so that W is the same function of $(\xi, \eta, \dots, \lambda)$ that V is of $(\alpha, \beta, \dots, \theta)$; say this is

$$W = -CH \int_{\lambda}^{\infty} \bar{T} \frac{t^{-q-1} dt}{\dots},$$

then we have for ρ the value

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q + 1) e^{2q+1}} \cdot \left[\left(1 - \frac{x^2}{f^2 + \lambda} - \dots - \frac{z^2}{h^2 + \lambda} \right) \cdot \frac{1}{\sqrt{f^2 + \lambda}} \frac{d}{d\xi} + \dots + \frac{1}{\sqrt{h^2 + \lambda}} \frac{d}{d\zeta} + \frac{1}{\sqrt{\lambda}} \frac{d}{d\iota} \right] W,$$

where e is to be put $= 0$.

126. Suppose e is $= 0$; then, if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} > 1$, λ is not $= 0$, but is the positive root of $\frac{x^2}{f^2 + \lambda} + \dots + \frac{z^2}{h^2 + \lambda} = 1$: $\iota = \sqrt{1 - \frac{x^2}{f^2 + \lambda} - \dots - \frac{z^2}{h^2 + \lambda}} = 0$; and we have $\rho = 0$, viz. ρ is $= 0$ for all points outside the ellipsoid $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1$.

But if $\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} < 1$, then, on writing $e = 0$, we have $\lambda = 0$, $\iota^{-1} = \frac{1}{\lambda}$,

$$\rho = \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q + 1)} \cdot \frac{e^{2q+1} \sqrt{\lambda}}{\dots} \left[\frac{x}{\sqrt{f^2 + \lambda}} \frac{1}{\sqrt{f^2 + \lambda}} \frac{d\bar{T}}{d\xi} + \dots + \frac{1}{\sqrt{h^2 + \lambda}} \cdot \frac{1}{\sqrt{h^2 + \lambda}} \frac{d\bar{T}}{d\zeta} + \frac{1}{\sqrt{\lambda}} \cdot \frac{1}{\sqrt{\lambda}} \frac{d\bar{T}}{d\iota} \right]_{\lambda=0},$$

where the term in () is

$$= -2\Theta\Lambda_0/(f \dots h) \cdot \lambda^{q+1}.$$

Hence

$$\begin{aligned} \rho &= \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q + 1) \Lambda_0 f g \dots h} \psi \\ &= \frac{\Gamma(\frac{1}{2}s + q)}{2(\sqrt{\pi})^s \Gamma(q + 1) \Lambda_0 f g \dots h} \psi\left(\frac{x}{f}, \frac{y}{g}, \dots, \frac{z}{h}\right), \end{aligned}$$

[p. 403] where ψ_0, Λ_0 are what ψ, Λ become on writing therein $\lambda = 0$. It will be remembered that Λ is what H becomes on changing therein θ into λ ; hence Λ_0 is what H becomes on writing therein $\theta = 0$.

Moreover ψ is what ϕ becomes on changing therein $\alpha, \beta, \dots, \gamma$ into $\xi, \eta, \dots, \zeta$: writing $\lambda = 0$, we have $\xi = \frac{x}{f}$, $\eta = \frac{y}{g}$, $\dots$, $\zeta = \frac{z}{h}$; hence ψ_0 is what ϕ becomes on changing therein $\alpha, \beta, \dots, \gamma$ into $\frac{x}{f}, \frac{y}{g}, \dots, \frac{z}{h}$.

And it is proper in ϕ to restore the original variables by writing $\frac{\alpha}{\sqrt{f^2 + \theta}}, \frac{\beta}{\sqrt{g^2 + \theta}}, \dots, \frac{\gamma}{\sqrt{h^2 + \theta}}$ in place of $\alpha, \beta, \dots, \gamma$.

127. Recapitulating,

$$V = \int \frac{\rho dx \dots dz}{\{(a - x)^2 + \dots + (c - z)^2 + e^2\}^{\frac{1}{2}s+q}},$$

where, since for the value of V about to be mentioned ρ vanishes for points outside the ellipsoid, the integral is to be taken over the ellipsoid

$$\frac{x^2}{f^2} + \dots + \frac{z^2}{h^2} = 1;$$

and then, transferring a constant factor, if

$$V = -\frac{\Gamma(\frac{1}{2}s + q)}{\Gamma(q + 1)} \Lambda_0(f \dots h) H \int_{\theta}^{\infty} \bar{T} \frac{t^{-q-1} dt}{\sqrt{(t + f^2) \dots (t + h^2)} \cdot \sqrt{t}},$$

the corresponding value of ρ is

$$\rho = \psi_0 \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2} \right)^q,$$

where Λ_0 is what H becomes on writing therein $\theta = 0$, and ψ_0 is what ψ becomes on writing $\frac{x}{f}, \frac{y}{g}, \dots, \frac{z}{h}$ in place of $\alpha, \beta, \dots, \gamma$.

128. Thus, putting for shortness $\bar{H} = t^{-q-1} \{(t+f^2) \dots (t+h^2)\}^{-\frac{1}{2}}$, we have in the three several cases $\phi = 1$, $\phi = \alpha$, $\phi = \alpha\beta$ respectively,

$$\begin{aligned} H = 1, \quad \rho &= \left(1 - \frac{x^2}{f^2} - \dots - \frac{z^2}{h^2}\right)^q, \quad V = \frac{\Gamma(\frac{1}{2}s+q)}{\Gamma(q+1)} (f \dots h) \int_{\theta}^{\infty} \bar{H}, \\ H &= \frac{\sqrt{f^2 + \theta}}{\sqrt{f^2 + g^2 + \dots + h^2}}, \quad \dots, \quad V = \dots \int_{\theta}^{\infty} \bar{H}, \\ H &= \frac{\sqrt{f^2 + \theta}}{\sqrt{f^2 + g^2 + \dots + h^2}} \cdot \frac{\sqrt{g^2 + \theta}}{\sqrt{f^2 + g^2 + \dots + h^2}}, \quad \dots, \quad V = \dots \int_{\theta}^{\infty} \frac{\bar{H} dt}{\sqrt{f^2 + t} \cdot \sqrt{g^2 + t}}. \end{aligned}$$

[p. 404] For the case last considered

$$\begin{aligned} \phi &= \frac{f^2 \alpha^2}{2q+2+\frac{1}{2}\kappa_0 f^2} + \dots + \frac{h^2 \gamma^2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}, \\ H &= \frac{f^2(f^2 + \theta)}{2q+2+\frac{1}{2}\kappa_0 f^2} + \dots + \frac{h^2(h^2 + \theta)}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}, \quad \bar{H} \text{ same function with } t \text{ for } \theta, \\ \psi_0 &= \frac{x^2}{2q+2+\frac{1}{2}\kappa_0 f^2} + \dots + \frac{z^2}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}, \\ \Lambda_0 &= \frac{f^4}{2q+2+\frac{1}{2}\kappa_0 f^2} + \dots + \frac{h^4}{2q+2+\frac{1}{2}\kappa_0 h^2} - \frac{1}{\kappa_0}, \end{aligned}$$

where κ_0 is the root of the equation

$$\frac{1}{2q+2+\frac{1}{2}\kappa_0 f^2} + \dots + \frac{1}{2q+2+\frac{1}{2}\kappa_0 h^2} + 1 = 0.$$

Annex VI. Examples of Theorem C. Art. Nos. 129 to 132.

129. First example: relating to the $(s+1)$ -coordinal sphere $a^2 + \dots + z^2 + w^2 = f^2$. Assume

$$V' = \frac{M}{(a^2 + \dots + c^2 + e^2)^{\frac{1}{2}(s-1)}}, \quad V'' = \frac{M}{f^{s-1}} \quad (\text{a constant});$$

these values each satisfy the potential equation.

V' is not infinite for any point outside the surface, and for indefinitely large distances it is of the proper form.

V'' is not infinite for any point inside the surface; and at the surface $V' = V''$. The conditions of the theorem are therefore satisfied. Writing

$$V = \int \frac{\rho dS}{\{(a-x)^2 + \dots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}(s-1)}},$$

we have

$$\rho = \frac{1}{4(\sqrt{\pi})^{s+1}} \left(\frac{dV'}{dn'} - \frac{dV''}{dn''} \right),$$

where

$$\frac{dV''}{dn''} = - \left(\frac{x}{f} \frac{d}{dx} + \dots + \frac{z}{f} \frac{d}{dz} + \frac{w}{f} \frac{d}{dw} \right) \frac{M}{(x^2 + \dots + z^2 + w^2)^{\frac{1}{2}(s-1)}} = \frac{(s-1)M(x^2 + \dots + z^2 + w^2)^{\frac{1}{2}(s-1)}}{f \cdot (x^2 + \dots + z^2 + w^2)^{\frac{1}{2}(s+1)}},$$

[p. 405] which at the surface is

$$= \frac{(s-1)M}{f^s}.$$

Hence

$$\rho = \frac{(s-1)\Gamma(\frac{1}{2}(s+1))M}{4(\sqrt{\pi})^{s+1}f^s} = \frac{\Gamma(\frac{1}{2}s+1)M}{2(\sqrt{\pi})^{s+1}f^s} \quad (\text{viz. } \rho \text{ is constant}).$$

130. Writing for convenience $M = \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))} \delta f^s S f^s$ (δf^s a constant which may be put = 1), also $a^2 + \dots + c^2 + e^2 = s^2$, we have $\rho = S f^s$, and consequently

$$\begin{aligned} \int \frac{\delta f^s dS}{\{(a-x)^2 + \dots + (c-z)^2 + (e-w)^2\}^{\frac{1}{2}(s-1)}} &= \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))} \cdot \frac{f^s}{s^{s-1}}, \quad \text{for exterior point } s > f, \\ &= \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))} \cdot f^{s-1}, \quad \text{for interior point } s < f. \end{aligned}$$

By making $a, \dots, c, e$ all indefinitely large, we find

$$\int \delta f^s dS = \frac{2(\sqrt{\pi})^{s+1}}{\Gamma(\frac{1}{2}(s+1))} f^s,$$

viz. the expression on the right-hand side is here the mass of the shell thickness δf .

Taking $s = 3$, we have the ordinary formulae for the Potential of a uniform spherical shell.

131. Suppose $s = 3$, but let the surface be the infinite cylinder $x^2 + y^2 = f^2$. Take here

$$V' = M \log \sqrt{a^2 + b^2}, \quad V'' = M \log f,$$

each satisfying the potential equation $\left(\frac{d^2V}{dx^2} + \frac{d^2V}{dy^2}\right) = 0$; but V , instead of vanishing, is infinite at infinity, and the conditions of the theorem are not satisfied; the Potential of the cylinder is in fact infinite. But the failure is a mere consequence of the special value of s , viz. this is such that $s - 2$, instead of being positive, is = 0. Reverting to the general case of $(s + 1)$ -dimensional space, let the surface be the infinite cylinder $x^2 + \dots + z^2 = f^2$; and assume

$$V' = \frac{M}{(a^2 + \dots + c^2)^{\frac{1}{2}(s-2)}}, \quad V'' = \frac{M}{f^{s-2}} \quad (\text{a constant}).$$

These satisfy the potential equation; viz. as regards V' , we have

[End of pages 381–405 (PDF pp. 401–425). Paper 607 continues in subsequent pages.]